

TwinHaus: Digital Twin Design Tool for Sustainable Building Construction

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Figure 1: Overview of the TwinHaus Interface: (Left) Augmented Reality view for structural overview. (Center) Virtual Reality in a suite view, visualizing energy and water consumption. (Right) Real-time environmental metrics on a wall-mounted dashboard.

ABSTRACT

As the construction industry increasingly prioritizes sustainability, the need for immersive tools that can visualize and analyze complex building performance factors has become essential. Many existing design solutions rely on static reports of simulations, often failing to convey the spatial and dynamic nature of airflow, water consumption, and heat transfer within spaces. We present TwinHaus, a unified digital twin framework that operates in Virtual Reality to make these invisible environmental phenomena perceptible in real time. The system integrates heuristic particle-based airflow simulation, fixture-linked water consumption visualization, and raycasting-driven thermodynamic modeling to reflect user interactions within a sustainable housing unit dynamically. A comprehensive environmental dashboard further connects these simulations to quantifiable metrics, allowing data-informed decisions to be made during the design process. [Click here for demo video.](#)

Index Terms: Virtual Reality, Augmented Reality, Digital Twin, Sustainable Design, Fluid Dynamics, Unity

1 INTRODUCTION

Sustainable building design is a critical component of modern construction, aiming to minimize environmental impact while maximizing occupant comfort [1, 7]. However, many key factors influencing building performance, such as airflow patterns, potential water usage rates, and thermal gradients, are invisible to the naked eye, making them difficult to conceptualize during the design phase [6, 8, 12]. Existing simulation tools often rely on 2D visualizations, limited to single-factor simulations, or on static charts, which may fail to convey the complexity of these phenomena [4].

TwinHaus aims to address this gap by providing a fully interactive digital twin that reveals invisible environmental phenomena in Augmented and Virtual Reality (AR and VR). Using the immersive capabilities, it creates a high-fidelity digital twin of a sustainable housing unit (modelled by professional architects), allowing users to step inside the design and interact with environmental variables in real time. We present details of the technical implementation, focusing on its core simulation modules: airflow dynamics, water flow visualization, and energy management.

2 RELATED WORK

Sustainable building design tools explore simulation-based decision support in the early design stages, focusing on energy use, comfort, and resource consumption [1, 2, 6]. The tools often use 2D plans, simplified dashboards, or static reports to communicate performance outcomes to designers [4, 7]. While such approaches can quantify environmental indicators, they frequently separate numerical results from spatial context, making it difficult to understand how airflow, thermal gradients, or water usage patterns unfold within specific rooms or circulation paths [8, 12]. This limitation is particularly evident when designers must compare multiple design options or assess local trade-offs between comfort and conservation.

Digital twins have been introduced in construction and facility management as a way to couple physical assets with virtual models that are continuously updated with data [3, 11]. There is a growing interest in using digital twins for sustainability, highlighting the potential of monitoring energy performance and environmental conditions throughout the building's life cycle [3]. However, many implementations emphasize backend data integration and analytics rather than interactive, in situ visualization of environmental processes during early design. Foundational work on digital twins frames them as a convergence of physical and virtual spaces to manage complex system behaviour, yet practical tools often provide limited support for intuitive, human-centred exploration of performance in the design phase [1, 11].

Existing work on real-time airflow visualization builds on fluid simulation methods that can be computationally intensive and are often used in offline workflows [9, 10]. Similarly, visualizations of thermal comfort or water usage are frequently presented as over-

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lays on 2D drawings or static 3D scenes rather than as interactive, physics-informed feedback inside an immersive model [4, 8].

3 SYSTEM OVERVIEW

TwinHaus provides a multi-scale interface for sustainable building analysis, operating in two complementary modes: an AR macro-view and a VR micro-view. The system is developed using Unity 6 to support high-fidelity rendering, particle-based simulation, and real-time environmental interaction. The system is optimized for the Meta Quest 3, maintaining a stable frame rate during particle-heavy simulations to ensure comfort. The AR module functions as a structural overview tool. By projecting the building model onto a physical surface, users are allowed to examine the full architectural layout, including internal structures from the foundation to the roof. This mode enables a rapid assessment of spatial organization and structural hierarchy before transforming to detailed interior analysis. The VR module immerses the user inside a furnished residential suite where environmental systems respond dynamically to user interaction. To support actionable decision-making, TwinHaus includes a real-time dashboard implemented as a user interface element mounted on an interior wall. The dashboard continuously updates as users interact with fixtures such as toggling lights, activating faucets, or altering ventilation to immediate numerical changes. The dashboard shows an intuitive understanding of resource efficacy. The interactions drive the real-time simulation of water, air, and thermal systems.

Water usage is represented through localized particle systems attached to fixtures. When activated, the system emits fluid particles to visualize flow rate while computing cumulative water consumption. The physics model tracks dispersed volume, providing immediate visual feedback on the resource intensity of each fixture.

Invisible air currents are simulated using ambient particle emitters governed by a custom `AirflowController`. To balance visual fidelity with the performance constraints of standalone XR hardware, TwinHaus utilizes a Hybrid Lagrangian-Eulerian framework. While high-fidelity Eulerian solvers utilize complex grid-based pressure projection, our system implements a computationally efficient Lagrangian particle-based heuristic. This module uses semi-implicit Euler integration to approximate natural convection, where thermal buoyancy influences vertical particle velocity (v_y), updated according to temperature (T):

$$v_y = v_{current} + (k \cdot \Delta t) \quad (1)$$

where $v_{current}$ is the previous-frame velocity, k is a temperature-dependent buoyancy constant (positive for $T > 25^\circ\text{C}$, negative for $T < 25^\circ\text{C}$), and Δt is the simulation timestep. To bridge the gap between simulation and quantifiable data, we implement fixed Eulerian observation zones using spherical triggers. These zones sample real-time particle data to compute the localized average air velocities displayed on the system dashboard. This enables users to visually identify and numerically verify warm air accumulation near ceilings and cold drafts near windows, providing an intuitive understanding of the building's aerodynamic performance.

The thermal behaviour of surfaces is modelled through a `LightBlockChecker` that computes heating from light sources and cooling from airflow. Heat absorption follows the standard specific heat relation [5]:

$$Q = mc\Delta T \quad (2)$$

where Q is absorbed energy, m is material mass, c is specific heat capacity, and ΔT is the temperature change. Real-time raycasting determines light exposure and occlusion, ensuring only illuminated surfaces undergo radiative heating. Cooling rates are modulated by airflow velocities produced by the convection simulation.

4 APPLICATION & LIMITATIONS

Users can identify poor air circulation through stagnation of particles and detect localized overheating. Particle-based simulations of airflow and water flow offer an intuitive understanding of resource usage, allowing designers to explore alternative layouts and immediately observe their effects. Real-time energy and water meters offer feedback, encouraging efficient resource usage. The graphical illustration may not be as accurate as the values measured directly from sensors, the physics model can provide a sufficient approximation for visualization and learning purposes.

5 CONCLUSION

TwinHaus illustrates the potential of digital twins for sustainable building design by making invisible environmental phenomena perceptible. Future work will incorporate real-world sensor data, enhance the physics engine to address complex fluid dynamics, and evaluate design effectiveness through user studies, providing a generalizable framework for interactive prefab utilization and informational digital twin applications.

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